

2b block by block: which parts are finite, and how to do them

Three of the six blocks need nothing at all; the other three carry one term of C_{UV} each

$$\eta = \frac{p^+}{k^+}, \quad \bar{\eta} = 1 - \eta, \quad L = \log \frac{k^+}{\Lambda}, \quad \mathcal{W} = \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2}, \quad C_{UV} = \eta \bar{\eta} + \frac{\eta}{\bar{\eta}} + \frac{\bar{\eta}}{\eta}$$

Yes — and the split is cleaner than you might expect. The six blocks of 2b divide into two groups that need completely different treatment:

- $T_3 + T_5 + T_6$: **nothing at all.** Their UV residues cancel *within the group* ($\frac{2}{d_\perp}(1 - \bar{\eta} - \eta) = 0$), and their weights $\bar{\eta}$, η , -1 are regular at both p^+ endpoints. No dimensional regularisation, no plus prescription — just integrate.
- T_1, T_2, T_4 : **one term of C_{UV} each.** Their UV residues are exactly $\eta \bar{\eta}$, $\frac{\eta}{\bar{\eta}}$, $\frac{\bar{\eta}}{\eta}$ — the three terms of C_{UV} , one per block. These need the counterterm. And T_2, T_4 are also the only two with p^+ poles, which is why the transverse subtraction must come **first**.

1. The block-by-block table

The UV residue of each block, as the coefficient of $\frac{\mathcal{W}}{k^+(z-x)^2}$, measured by exact angular quadrature at $|z - x| = 10^{-4}$:

block	p^+ weight	UV residue	$d_\perp = 2$	$d_\perp = 3$	$d_\perp = 4$	p^+ endpoints
T_1	$\eta \bar{\eta}$	$\eta \bar{\eta}$	0.22750000	0.22749996	0.22750000	regular
T_2	$-\frac{\eta}{\bar{\eta}}$	$\frac{\eta}{\bar{\eta}}$	0.53846154	0.53846154	0.53846154	pole at $p^+ = k^+$
T_3	$\bar{\eta}$	$-\frac{2\bar{\eta}}{d_\perp}$	-0.65000001	-0.43333325	-0.32499999	regular
T_4	$-\frac{\bar{\eta}}{\eta}$	$\frac{\bar{\eta}}{\eta}$	1.85714285	1.85714285	1.85714285	pole at $p^+ = 0$
T_5	η	$-\frac{2\eta}{d_\perp}$	-0.35000001	-0.23333329	-0.17500000	regular
T_6	-1	$+\frac{2}{d_\perp}$	1.00000002	0.66666655	0.49999999	regular
$T_3 + T_5 + T_6$			4×10^{-9}	3×10^{-9}	3×10^{-9}	UV finite
$T_1 + T_2 + T_4$			2.62310439	2.62310435	2.62310439	$= C_{UV} = 2.62310440$

(at $\eta = 0.35$; the residues of T_1, T_2, T_4 carry no $d_\perp$ at all, while those of T_3, T_5, T_6 are pure $\frac{1}{d_\perp}$ and cancel because $\frac{k^+}{p^+} - \frac{k^+}{q} = 0$.)

Keep T_3, T_5, T_6 together. Individually each one is log divergent at $z \rightarrow x$ — it is only the sum that is finite. If you split them across different equations you will hand yourself three spurious poles that have to cancel later.

2. Group (B): $T_3 + T_5 + T_6$ — nothing special

$$\left. \frac{d^3 N_{2b}^{\text{NLO}}}{d^3 k} \right|_{(B)} = \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{1}{k^+} \int_\Lambda^{k^+ - \Lambda} \frac{dp^+}{2\pi} \int_{x, y, z, x'} e^{-ik \cdot (w' - \bar{\eta}z - \eta x)} \frac{(x' - w')^i (z - x)^m}{(x' - w')^2 (z - x)^2} \left[T_3 + T_5 + T_6 \right]_{im} \times (\text{colour})$$

with everything at $d_\perp = 2$ and no prescription anywhere:

- $z \rightarrow x$: finite, by the cancellation above.
- Large $|z|$: cut by the phase $e^{i\bar{\eta}k \cdot z}$ at $|z| \sim \frac{1}{\bar{\eta}k_\perp}$.
- $p^+ \rightarrow 0$ and $p^+ \rightarrow k^+$: the weights $\bar{\eta}$, η , -1 are bounded and the companion $h(\eta) = \frac{e^{-i\eta\kappa}}{\eta A + \bar{\eta}B}$ is smooth, so the dp^+ integral simply converges. You may even extend it to $\int_0^{k^+}$ — the difference is $O(\Lambda)$.

3. Group (A): T_1, T_2, T_4 — subtract first, then p^+

Because T_2 and T_4 carry *both* a UV divergence and a p^+ pole, the order is forced. Subtract block by block, each with its own weight:

$$\mathcal{S}_n(\eta) = w_n(\eta) \mathcal{W} \frac{e^{i\bar{\eta} k \cdot (z-x)}}{k^+ (z-x)^2}, \quad w_1 = \eta \bar{\eta}, \quad w_2 = \frac{\eta}{\bar{\eta}}, \quad w_4 = \frac{\bar{\eta}}{\eta}.$$

(i) The counterterms — closed form, no prescription needed

Each integrates in z to $-\pi \left[\frac{1}{\epsilon} + \log \frac{\bar{\eta}^2 k^2}{\bar{\mu}^2} \right]$, and the two η -moments are elementary:

block	$\int w d\eta$	quadrature	$\int w \log \bar{\eta}^2 d\eta$	quadrature
T_1	$\frac{1}{6}$	0.16666667	$-\frac{5}{18}$	-0.27777778
T_2	$L - 1$	17.4206808	$-L^2 + 2$	-337.321479
T_4	$L - 1$	17.4206808	$2 - \frac{\pi^2}{3}$	-1.28986811
sum	$2L - \frac{11}{6}$		$-L^2 + \frac{67}{18} - \frac{\pi^2}{3}$	

(quadrature at $\Lambda/k^+ = 10^{-8}$, matching the closed forms to 8 digits). So each block's UV piece is

$$-\frac{1}{(2\pi)^3} \frac{g^4}{16\pi^4} \left\{ \left(\int w_n \right) \left[\frac{1}{\epsilon} + \log \frac{k^2}{\bar{\mu}^2} \right] + \left(\int w_n \log \bar{\eta}^2 \right) \right\} \times \int_{x,y,x',w'} e^{-ik \cdot (w'-x)} \mathcal{W} \times (\text{colour}),$$

and the three add to the I_0, I_1 of the previous note. Notice where the $-\frac{11}{6}$ comes from: T_1 contributes $+\frac{1}{6}$ and T_2, T_4 contribute -1 each.

(ii) The remainders — now the plus prescription is allowed

Check first that the subtraction really cured the endpoint residues. Each residue on its own is a constant in $|z-x|$ (i.e. log divergent, value 0.0322 in the test configuration); the difference falls as $|z-x|^2$:

block	residue	counterterm	difference at $ z-x = 10^{-3}$	10^{-4}	10^{-5}
T_4 at $\eta \rightarrow 0$	0.03223204	0.03223204	-2.0×10^{-9}	-2.0×10^{-11}	-2.0×10^{-13}
T_2 at $\eta \rightarrow 1$	0.03223204	0.03223204	-9.5×10^{-15}	-9.4×10^{-17}	$\sim 10^{-18}$

So the remainders can be plus-prescribed, **one-sided each**:

$$T_4 \text{ remainder : } \left[\frac{1}{\eta} \right]_+ \text{ at } p^+ \rightarrow 0, \quad T_2 \text{ remainder : } \left[\frac{1}{1-\eta} \right]_+ \text{ at } p^+ \rightarrow k^+,$$

each producing one δ term with $\log \frac{k^+}{\Lambda}$, while T_1 's remainder has no endpoint pole at all and is an ordinary convergent double integral.

Do not plus-prescribe before subtracting. Before the subtraction, $g(\text{endpoint})$ for T_2 and T_4 is a UV-divergent transverse integral, and $\int \frac{dn}{\eta} [g(\eta) - g(0)]$ requires $g(0)$ finite. You would be subtracting infinity from infinity and the finite part would be scheme-dependent garbage. This is the one place in $2b$ where the order of operations actually matters.

4. The recipe, in order

1. Write $2b = (A) + (B)$ with $(A) = T_1 + T_2 + T_4$ and $(B) = T_3 + T_5 + T_6$.
2. **(B)**: set $d_\perp = 2$, integrate dp^+ directly, done. No regulator, no prescription.
3. **(A)**: subtract $\mathcal{S}_1 + \mathcal{S}_2 + \mathcal{S}_4$ from the integrand.
4. The subtracted counterterms: use the closed forms in the table of §3(i). This is where the $\frac{1}{\epsilon}$, the $\log \frac{k_\perp^2}{\bar{\mu}^2}$, the $-\frac{11}{6}$ (β_0) and the $-L^2$ live.
5. The remainders: $d_\perp = 2$, then one-sided plus prescriptions on T_2 and T_4 , nothing on T_1 .

Net: **one** $\frac{1}{\epsilon}$ in all of $2b$, four $\delta(1 - \dots)$ -type $\log \frac{k^+}{\Lambda}$ terms (two from the counterterms' $2L$, two from the remainders' plus prescriptions), and three finite multi-dimensional integrals that are done numerically.

Checks: `src/rowB2_2b_blocks.py` — §1 the per-block UV residues by exact angular quadrature in $d_\perp = 2, 3, 4$ (the leading residue is a quadratic form in $\hat{u}$, so averaging over the $2d_\perp$ directions $\pm e_i$ is exact); §3(i) the two η -moments per block against mpmath quadrature; §3(ii) the subtracted endpoint residues. Output in `RESULTS_rowB2_2b_blocks.txt`. The overall $2b$ result is in `src/rowB2_2b_uv.py`.